UNIVERSITY OF DERBY

College of Science and Engineering

Mock Paper 2324 Solutions

OPERATIONAL RESEARCH – 5CM522

1. Forecasting

(a) Using periods 1–12, the centred 4-period moving averages (CMA) are:

Period	Sales	4-MA	CMA
3	1002	1015.25	1015.625
4	1021	1025.25	1024.625
5	1053	1037.25	1037.125
6	1067	1050.50	1050.750
7	1042	1065.50	1065.875
8	1060	1078.00	1078.875
9	1093	1090.75	1091.375
10	1106	1100.75	1101.000

Fitting a straight line to the CMA values gives the trend

$$\hat{T}_t = 990.442 + 10.009t$$

(b) Seasonal indices from periods 1–12:

Season	Additive Index	Multiplicative Index
Spring	12.438	1.01187
Summer	15.875	1.01498
Autumn	-18.375	0.98247
Winter	-9.938	0.99068

The additive indices sum to 0 and the multiplicative indices average to 1, so they are correctly normalised.

(c) Validation forecasts for periods 13–16:

Period	Actual	Additive	Multiplicative	Additive AE	Multiplicative AE
13	1134	1132.996	1133.857	1.004	0.143
14	1147	1146.442	1147.505	0.558	0.505
15	1123	1122.201	1120.581	0.799	2.419
16	1139	1140.647	1139.862	1.647	0.862

Using mean absolute deviation:

$$\text{MAD}_{\text{add}} = 1.002$$

$$\text{MAD}_{\text{mult}} = 0.982$$

So the multiplicative model is marginally better on the validation set.

(d) Forecasts for periods 17–20 using the multiplicative model:

Period	Season	Forecast
17	Spring	1174.367
18	Summer	1188.140
19	Autumn	1159.915
20	Winter	1179.525

The multiplicative model is chosen because it gives the lower validation MAD, although the difference is small.

2. Simulation

(a) Stochastic inputs: weekly deliveries and weekly demand.

State variable: stock remaining at the end of the week, which then becomes part of next week's starting stock.

Measures: weekly total cost, average weekly cost, overflow usage and any missed sales.

(b) Capacity 9 simulation:

Week	Delivered	Demand	Start Stock	Sold	End Stock	Overflow	Total Cost
1	7	8	7	7	0	0	90
2	8	9	8	8	0	0	90
3	10	10	10	10	0	1	110
4	9	11	9	9	0	0	90
5	7	8	7	7	0	0	90
6	9	9	9	9	0	0	90
7	9	10	9	9	0	0	90
8	8	8	8	8	0	0	90
9	9	6	9	6	3	0	90
10	8	11	11	11	0	2	130
11	9	8	9	8	1	0	90
12	9	8	10	8	2	1	110

Weekly cost formula:

$$\text{Cost} = 10(9) + 20\max(\text{start stock} - 9, 0)$$

(c) Average weekly cost for capacity 9:

$$\frac{90 + 90 + 110 + 90 + 90 + 90 + 90 + 90 + 90 + 130 + 90 + 110}{12} = 96.667$$

So the estimated average weekly cost is £96.67.

(d) Comparison of capacities:

Capacity	Average Weekly Cost (£)
7	106.67
8	100.00
9	96.67
10	101.67

Among these four options, capacity 9 gives the lowest average weekly cost.

(e) A manager should be cautious because:

- one short run may be unrepresentative due to random variation;
- the result depends on one particular set of random numbers;
- longer runs or repeated replications would give a more reliable estimate of average cost.

3. Transportation / Transhipment

(a) Let x_{ij} be the amount sent from warehouse i to shop j.

$$\min Z = 8x_{AX} + 7x_{AY} + 5x_{AZ} + 12x_{BX} + 3x_{BY} + 9x_{BZ} + 10x_{CX} + 11x_{CY} + 7x_{CZ}$$

subject to supply constraints

$$x_{AX} + x_{AY} + x_{AZ} \leq 100$$

$$x_{BX} + x_{BY} + x_{BZ} \leq 100$$

$$x_{CX} + x_{CY} + x_{CZ} \leq 100$$

demand constraints

$$x_{AX} + x_{BX} + x_{CX} = 70$$

$$x_{AY} + x_{BY} + x_{CY} = 80$$

$$x_{AZ} + x_{BZ} + x_{CZ} = 150$$

$$x_{ij} \geq 0$$

(b) One optimal direct-shipping plan is:

Route	Shipment	Cost Contribution
$A \to X$	50	400
$A \to Z$	50	250
$B \to X$	20	240
$B \to Y$	80	240
$C \to Z$	100	700

Minimum total transport cost:

$$400 + 250 + 240 + 240 + 700 = 1830$$

(c) With $x_{BY} = 0$ and $x_{AZ} \geq 5$, one optimal plan is:

Route	Shipment
$A \to X$	15
$A \to Y$	80
$A \to Z$	5
$B \to Z$	100
$C \to X$	55
$C \to Z$	45

Revised total cost:

$$15(8) + 80(7) + 5(5) + 100(9) + 55(10) + 45(7) = 2470$$

(d) For the transhipment model, let x_{ik} be flow from warehouse i to hub k and y_{kj} be flow from hub k to shop j.

Add:

$$\sum_i x_{ik} = \sum_j y_{kj}$$

for each hub k.

An optimal transhipment plan is:

Route	Shipment
$A \to D_1$	100
$B \to D_1$	50
$B \to D_2$	50
$C \to D_2$	100
$D_1 \to Z$	150
$D_2 \to X$	70
$D_2 \to Y$	80

So both intermediary points are used.

Total transhipment cost:

$$100(4) + 50(3) + 50(2) + 100(4) + 150(2) + 70(2) + 80(4) = 1810$$

This is lower than the direct-shipping optimum, so the intermediary-point option is worthwhile.

4. Linear Programming

(a) Let x be weekly production of product X and y be weekly production of product Y.

$$\max Z = 36x + 34y$$

subject to

$$4x + 3y \leq 40$$

$$0.2x + 0.3y \leq 3$$

$$x, y \geq 0$$

(b) At the optimum both resource constraints are binding:

$$4x + 3y = 40, \qquad 0.2x + 0.3y = 3$$

Multiplying the second equation by 10 gives

$$2x + 3y = 30$$

Subtracting gives

$$2x = 10 \Rightarrow x = 5$$

Then

$$2(5) + 3y = 30 \Rightarrow 3y = 20 \Rightarrow y = \frac{20}{3}$$

Maximum profit:

$$Z = 36(5) + 34\left(\frac{20}{3}\right) = \frac{1220}{3} \approx 406.67$$

(c) Both raw material constraints are binding, so unused material is:

$$\text{raw material } A : 0$$

$$\text{raw material } B : 0$$

(d) The extra condition is

$$y = 3x$$

So the revised model is

$$\max Z = 36x + 34y$$

subject to

$$4x + 3y \leq 40$$

$$0.2x + 0.3y \leq 3$$

$$y = 3x$$

$$x, y \geq 0$$

(e) Substituting $y = 3x$:

$$4x + 9x \leq 40 \Rightarrow 13x \leq 40$$

$$0.2x + 0.9x \leq 3 \Rightarrow 1.1x \leq 3$$

So

$$x \leq \frac{30}{11}$$

is the tighter condition, hence

$$x = \frac{30}{11}, \qquad y = \frac{90}{11}$$

Revised profit:

$$Z = 36\left(\frac{30}{11}\right) + 34\left(\frac{90}{11}\right) = \frac{4140}{11} \approx 376.36$$

So the linked-production rule reduces profit by

$$406.67 - 376.36 \approx 30.30$$